

● Project Overview

A multilabel deep-learning system that classifies user comments into six toxicity categories (toxic, severe_toxic, obscene, threat, insult, identity_hate) using a CNN-GRU hybrid architecture with a custom weighted binary cross-entropy loss.

● Prerequisites

- > Python 3.8+ with pip installed
- > GPU recommended (CUDA-compatible) but not required — CPU training will be slower
- > ~2 GB free disk space (dataset + saved models)

● Installation

Install all required dependencies:

```
pip install pandas numpy matplotlib seaborn wordcloud nltk
pip install tensorflow keras scikit-learn
pip install keras-preprocessing pyarrow
```

Then download NLTK stopwords (runs automatically in the notebook, or manually):

```
python -c "import nltk; nltk.download('stopwords')"
```

● Running the Notebook

1. Launch Jupyter and open the notebook:

```
jupyter notebook FILTRO_ANTI_HATER_CONSEGNAMI__1_.ipynb
```

2. Run all cells sequentially (Cell > Run All). The notebook will automatically download the dataset from the S3 bucket — no manual download needed.
3. The preprocessing step takes several minutes. After the first run, a `preprocessed_data.parquet` file is saved locally; subsequent runs load it instantly.
4. Trained models are saved to `models/` and training histories to `histories/`. If these folders already exist with saved files, you can skip training cells and load directly.

● Project Structure

```
.
|- FILTRO_ANTI_HATER_CONSEGNAMI__1_.ipynb    # Main notebook
|- preprocessed_data.parquet                  # Cached preprocessed data
|- models/                                   # Saved Keras models (.keras)
|   |- weighted_rnn_model.keras
|   |- GRU_rnn_model.keras
|   |- GRU_model.keras                      # Bidirectional GRU
|   |- CNN_RNN_model.keras
|   |- final_model.keras                    # Best performing model
|- histories/                                # Pickled training histories (.pkl)
```

● Key Notes

- > The notebook trains 5 deep-learning models sequentially — expect ~30-60 min total on GPU.
- > EarlyStopping (patience=4) halts training automatically if validation loss plateaus.
- > To skip to the best model only: run cells up to preprocessing, then jump to 'Final Model'.